

USB Sentinel

A professional USB diagnostic, monitoring, and analysis platform.

Project Vision

USB Sentinel is a cross-platform USB diagnostic and monitoring application designed for hardware developers, IT professionals, support engineers, and power users.

Unlike traditional USB viewers that simply enumerate connected devices, USB Sentinel is designed to become a complete USB analysis platform capable of monitoring, recording, comparing, and diagnosing USB devices over time.

The project originated while troubleshooting a Tobii PCEye Plus eye tracker but is intentionally designed to support **all USB devices**, regardless of manufacturer.

The long-term objective is to build an extensible platform that combines deep USB inspection, Windows device diagnostics, historical analysis, intelligent reporting, and both command-line and graphical interfaces into a single application.

Project Goals

USB Sentinel should:

- Detect and enumerate every USB device attached to the system.
 - Decode USB descriptors into human-readable information.
 - Record complete scan history.
 - Compare historical scans.
 - Detect changes automatically.
 - Diagnose common USB problems.
 - Generate professional reports.
 - Support vendor-specific diagnostic plugins.
 - Provide both an advanced CLI and a modern GUI.
 - Be useful for casual users while exposing every available detail to power users.
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Design Philosophy

The project follows several guiding principles:

Engine First

The backend engine is the heart of the application.

All USB scanning, diagnostics, logging, reporting, database access, and analysis are performed by the engine.

The CLI and GUI are simply different interfaces built on top of the same backend.

This guarantees:

- one source of truth
 - no duplicated logic
 - easier testing
 - easier maintenance
 - future web/API support
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Modular Architecture

Every major feature lives in its own module.

Scanner

History

Reports

Diagnostics

Plugins

GUI

CLI

Vendor Database

Each module should have a single responsibility.

Extensibility

USB Sentinel is intended to become a platform rather than a single application.

Support for vendors, protocols, report formats, and operating systems should be added through plugins whenever practical.

Human-Friendly

Raw USB descriptors are valuable.

Human-readable explanations are even better.

Whenever possible the software should translate hexadecimal values into meaningful descriptions.

Example:

Instead of

Class 03

display

Human Interface Device (HID)

Major Features

USB Enumeration

Enumerate every connected USB device.

Collect:

- Vendor ID
- Product ID
- Manufacturer
- Product
- Serial Number
- USB Version
- Device Class
- Device Subclass
- Device Protocol
- Bus Address
- Configurations
- Interfaces
- Alternate Settings

- Endpoints
 - Power Requirements
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USB Topology

Display the complete USB hierarchy.

Example

Computer

USB Root Hub

|—— Hub

| |—— Keyboard

| |—— Mouse

| |—— Camera

|—— Bluetooth Adapter

Vendor Identification

The application ships with a built-in vendor dictionary covering the most common USB vendors.

Examples include:

- Microsoft
- Logitech
- Intel
- Apple
- Tobii
- FTDI
- Arduino
- Raspberry Pi Foundation

Unknown Vendor IDs are automatically resolved using the official USB-IF vendor database and cached locally.

Logging

Every scan is permanently recorded.

Supported formats:

- TXT
- CSV
- JSON
- HTML

Future support:

- PDF

Each log entry contains:

- Timestamp
- Computer information
- Operating system
- Python version
- USB library versions
- Complete scan
- Errors encountered

Logs are append-only.

Nothing is overwritten.

History Database

USB Sentinel maintains a SQLite database containing historical information.

Each device record stores:

- First Seen
- Last Seen
- Detection Count
- Serial Number
- Driver History
- Interface History
- Endpoint History

This allows long-term monitoring of intermittent hardware problems.

Scan Comparison

Two scans can be compared automatically.

Changes detected include:

- New devices
- Removed devices
- Driver updates
- Firmware changes
- Configuration changes
- Interface changes
- Endpoint changes

Future versions will include graphical side-by-side comparisons.

Windows Diagnostics

Windows-specific information includes:

- Device Manager status
- Driver provider
- Driver version
- INF file
- Hardware IDs
- Compatible IDs
- Power management
- Device problem codes

Future versions will include:

- Kernel-PnP event analysis
 - USB event history
 - Driver installation history
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Plugin System

Vendor-specific plugins extend the core engine.

Examples:

- Tobii
- Logitech
- FTDI

- Arduino
- HID
- USB Cameras

Plugins may contribute:

- device recognition
- firmware validation
- driver recommendations
- troubleshooting advice
- health analysis

Command-Line Interface

Designed for power users.

Example commands:

scan

history

diff

report

topology

live

Common options:

--help

--version

--vid

--pid

--manufacturer

--serial

--driver

--interfaces

--endpoints

--tree

--changed

--new

--removed

--verbose

--json

--csv

--text

--all

--gui

Graphical User Interface

The GUI is intended to provide a modern desktop experience.

Design goals:

- Dark theme
- Fast
- Responsive
- Searchable
- Dockable panels
- Large data tables
- Tree navigation

Main sections:

- Device Tree
- Device Details
- History

- Events
 - Driver Information
 - Raw USB Descriptors
 - Scan Comparison
 - Reports
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Live Monitoring

Monitor USB activity in real time.

Detect:

- Device insertion
- Device removal
- Driver changes
- Configuration changes

Events are immediately recorded.

Intelligent Analysis

USB Sentinel should eventually move beyond displaying information.

It should analyze devices and identify likely problems.

Examples:

- Missing interfaces
- Incorrect drivers
- Endpoint inconsistencies
- Repeated disconnects
- Power issues
- Firmware mismatches

The software should provide probable causes along with recommended actions.

USB Health Score

Long-term goal:

Assign every device a health score.

Example:

USB Health

93 / 100

Warnings:

- Device disconnected 17 times during the past week.
- Driver updated yesterday.
- Expected interface missing.
- Device operating below expected USB speed.

Recommendations should accompany every warning.

Technical Stack

Language

- Python 3.12+

Libraries

- PyUSB
- libusb_package
- PySide6 (GUI)

Database

- SQLite

Reports

- JSON
- CSV
- HTML
- TXT

Packaging

- PyInstaller

Testing

- pytest
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Development Roadmap

Phase 0

Project setup

Architecture

Repository

Directory layout

Coding standards

Phase 1

USB scanner

Phase 2

Deep descriptor enumeration

Phase 3

Logging

Phase 4

Vendor database

Phase 5

History database

Phase 6

Scan comparison

Phase 7

Windows diagnostics

Phase 8

Command-line interface

Phase 9

USB topology

Phase 10

Plugin architecture

Phase 11

Live monitoring

Phase 12

Graphical user interface

Phase 13

Reporting engine

Phase 14

Intelligent analysis

Phase 15

Release candidate and production polish

Long-Term Vision

USB Sentinel is intended to become a comprehensive USB diagnostic platform rather than a simple device viewer.

The ultimate goal is to provide a tool that enables users to understand not only **what USB devices are connected**, but also **how they are functioning, how they have changed over time**, and **why problems occur**, all while presenting the information in a professional, approachable, and extensible interface.